Question

Ozone is an important air pollutant in the lower atmosphere, capable of triggering photochemical smog, among other effects. In alkaline aqueous solutions, ozone can decompose, while the small amount of H_2O_2 produced during decomposition also catalyzes ozone degradation. A possible mechanistic scheme is shown below. In alkaline solutions, HO_2^- can be considered fully ionized, and the ionization constant of water is K_w.

$$\mathrm{O_3 + OH^- \rightarrow HO_2^- + O_2} \quad k_1$$

$$\mathrm{H_2O_2 = HO_2^- + H^+} \quad K_2$$

$$\mathrm{O_3 + HO_2^- \rightarrow HO\cdot + O_2^- + O_2} \quad k_3$$

$$\mathrm{O_3 + HO\cdot \rightarrow HO_2\cdot + O_2} \quad k_4$$

$$\mathrm{H_2O_2 + HO\cdot \rightarrow HO_2\cdot + H_2O} \quad k_5$$

$$\mathrm{HO_2^- + HO\cdot \rightarrow HO_2\cdot + OH^-} \quad k_6$$

$$\mathrm{HO_2\cdot = H^+ + O_2^-} \quad K_7$$

$$\mathrm{OH^- + OH\cdot \rightarrow H_2O + O^-} \quad k_8$$

$$O_3 + O_2^- \rightarrow O_3^- + O_2 \quad k_9$$

$$O_3^- + H^+ \rightarrow HO\cdot + O_2 \quad k_{10}$$

$$O^- + H_2O \rightarrow HO\cdot + HO^- \quad k_{11}$$

Assuming that the chain termination step is solely the dimerization of hydroxyl radicals $OH\cdot$ (with a rate constant of k_{12}), and considering the chain initiation rate, derive the expression for $[O_2^-]$ in terms of the concentrations of $[H_2O_2]$, $[OH^-]$, and $[O_3]$, using reasonable steady-state approximations. Additionally, analyze v_{O_3}, the rate of ozone consumption, and select the most appropriate option from the choices below.

A. Expand the original expression completely into a sum of single terms with addition and subtraction operations (excluding the operations inside the square root), where each single term is the product of the coefficient, the power of the concentration term, and the operation inside the square root (if any). The expression for $[O_2^-]$ includes 1 term, and the expression for v_{O_3} includes 4 terms.

B. Expand the original expression completely into a sum of monomials involving addition and subtraction operations (excluding the content inside the square root), where each monomial is the product of a coefficient and the power of a concentration term multiplied by the operation inside the square root (if any). The expression for $[O_2^-]$ includes 1 term, and the expression for v_{O_3} includes 3 terms.

C. Expand the original expression completely into the sum of monomials involving addition and subtraction operations (excluding the content inside the square root), where each monomial is the product of a coefficient, a power term of concentration, and the operation inside the square root (if any). The expression for $[O_2^-]$ includes 2 terms, and the expression for v_{O_3} includes 3 terms.

D. Expand the original expression completely into a sum of individual terms with addition and subtraction operations (excluding the contents inside the square root). Each individual term is the product of a coefficient, a power of the concentration term, and the operation inside the square root (if any). The $[O_2^-]$ expression includes 2 terms, and the v_{O_3} expression includes 2 terms.

E. Expand the original expression completely into the sum of monomials involving addition and subtraction operations (excluding the content inside the square root), where each monomial is the product of a coefficient, the power of the concentration term, and the operation inside the square root (if any). The expression for $[O_2^-]$ includes 3 terms, and the expression for v_{O_3} includes 2 terms.

F. Expand the original expression completely into a sum of monomials involving addition and subtraction operations (excluding the operations inside the square root), where each monomial is the product of a coefficient, the power of a concentration term, and the operation inside the square root (if applicable). The expression for $[O_2^-]$ includes 3 terms, and the expression for v_{O_3} includes 1 term.

G. Fully expand the original expression into the sum of single terms with addition and subtraction operations (excluding the operations inside the square root), where each single term is the product of the coefficient, the power of the concentration term, and the operation inside the square root (if any). The $[O_2^-]$ expression includes 4 terms, and the v_{O_3} expression includes 1 term.

H. Expand the original expression completely into a sum of single terms with addition and subtraction operations (excluding the operations inside the square root), where each single term is the product of a coefficient and the power of the concentration term multiplied by the operation inside the square root (if any). The expression for $[O_2^-]$ includes 4 terms, and the expression for v_{O_3} includes 4 terms.

I. Expand the original expression fully into the sum of monomials involving addition and subtraction (excluding operations inside the square root), where each monomial is the product of a coefficient, the power of concentration terms, and the operation inside the square root (if any). The sum of the integer coefficients in the $[O_2^-]$ expression that do not include rate constants or equilibrium constants is 1, and the sum of the integer coefficients in the v_{O_3} expression that do not include rate constants or equilibrium constants is 10.

J. Fully expand the original expression into the sum of individual terms involving addition and subtraction operations (excluding operations inside the square root), where each term is the product of a coefficient, the power of the concentration term, and the operation inside the square root (if present). The sum of the integer coefficients in the $[O_2^-]$ expression that do not include rate constants or equilibrium constants is 2, and the sum of the integer coefficients in the v_{O_3} expression that do not include rate constants or equilibrium constants is 9.

K. Expand the original expression completely into a sum of monomials with addition and subtraction operations (excluding operations inside the square root), where each monomial is the product of a coefficient, power terms of concentration terms, and operations inside the square root (if any). The sum of the integer coefficients in the $[O_2^-]$ expression that do not include rate constants or equilibrium constants is 3, and the sum of the integer coefficients in the v_{O_3} expression that do not include rate constants or equilibrium constants is 8.

L. Fully expand the original expression into a sum of monomials with addition and subtraction operations (excluding operations inside square roots), where each monomial is the product of a coefficient, power terms of concentration terms, and operations inside square roots (if any). The sum of integer coefficients in the $[O_2^-]$ expression that do not include rate constants or equilibrium constants is 4, and the sum of integer coefficients in the v_{O_3} expression that do not include rate constants or equilibrium constants is 7.

M. Fully expand the original expression into the sum of monomials involving addition and subtraction operations (excluding operations inside the square root), where each monomial is the product of a coefficient and the power of concentration terms multiplied by the operations inside the square root (if any). The sum of integer coefficients in the $[O_2^-]$ expression that do not include rate constants or equilibrium constants is 5, and the sum of integer coefficients in the v_{O_3} expression that do not include rate constants or equilibrium constants is 6.

N. Expand the original expression completely into the sum of monomials with addition and subtraction operations (excluding the operations inside the square root), where each monomial is the product of a coefficient and the power of the concentration term and the operation inside the square root (if any). The sum of the integer coefficients in the $[O_2^-]$ expression that do not include rate constants or equilibrium constants is 6, and the sum of the integer coefficients in the v_{O_3} expression that do not include rate constants or equilibrium constants is 5.

O. Expand the original expression completely into the sum of monomials involving addition and subtraction operations (excluding the operations inside the square root), where each monomial is the product of a coefficient, the power of a concentration term, and the operation inside the square root (if any). The sum of the integer coefficients in the $[O_2^-]$ expression that do not include rate constants or equilibrium constants is 7, and the sum of the integer coefficients in the v_{O_3} expression that do not include rate constants or equilibrium constants is 4.

P. Expand the original expression completely into the sum of single terms with addition and subtraction operations (excluding the operations inside the square root), where each single term is the product of a coefficient, the power of the concentration term, and the operation inside the square root (if any). The sum of the integer coefficients in the $[O_2^-]$ expression that do not include rate constants or equilibrium constants is 8, and the sum of the integer coefficients in the v_{O_3} expression that do not include rate constants or equilibrium constants is 3.

Q. Expand the original expression completely into the sum of individual terms involving addition and subtraction (excluding operations inside the square root), where each term is the product of a coefficient and the power of a concentration term multiplied by the operation inside the square root (if any). The sum of the integer coefficients in the $[O_2^-]$ expression that do not include rate constants or equilibrium constants is 9, and the sum of the integer coefficients in the v_{O_3} expression that do not include rate constants or equilibrium constants is 2.

R. Expand the original expression completely into the sum of monomials involving addition and subtraction operations (excluding the operations inside the square root), where each monomial is the product of a coefficient and the power of concentration terms multiplied by the operations inside the square root (if any). The sum of the integer coefficients in the $[O_2^-]$ expression that do not include rate constants or equilibrium constants is 10, and the sum of the integer coefficients in the v_{O_3} expression that do not include rate constants or equilibrium constants is 1.

S. Due to certain reasons, options A-R are all correct

T. For certain reasons, options A-R are all incorrect.

U. Options A-R have more than 3 correct answers

V. Option A-R has exactly 2 correct.

Answer

Correct Answer: L

Detailed Explanation

According to the problem, the chain termination step is solely the dimerization of hydroxyl radicals, i.e.:

$$2\mathrm{HO}\cdot \rightarrow \mathrm{H_2O_2} \quad k_{12}$$

For other intermediate products (radicals and reactive intermediates), we can apply the steady-state approximation, meaning their generation rate equals their consumption rate.

The steady-state approximation yields:

$$d[\mathrm{HO}\cdot]/dt = k_3[\mathrm{O_3}][\mathrm{HO_2^-}] + k_{10}[\mathrm{O_3^-}][\mathrm{H^+}] + k_{11}[\mathrm{O^-}][\mathrm{H_2O}] - k_4[\mathrm{O_3}][\mathrm{HO}\cdot] - k_5[\mathrm{H_2O_2}][\mathrm{HO}\cdot] - k_6[\mathrm{HO_2^-}][\mathrm{HO}\cdot] - k_8[\mathrm{OH^-}][\mathrm{HO}\cdot] - 2k_{12}[\mathrm{HO}\cdot]^2 = 0$$

$$d[\mathrm{HO_2}\cdot]/dt = k_4[\mathrm{O_3}][\mathrm{HO}\cdot] + k_5[\mathrm{H_2O_2}][\mathrm{HO}\cdot] + k_6[\mathrm{HO_2^-}][\mathrm{HO}\cdot] - k_7[\mathrm{HO_2}\cdot] + k_{-7}[\mathrm{H^+}][\mathrm{O_2^-}] = 0$$

$$d[\mathrm{O_2^-}]/dt = k_3[\mathrm{O_3}][\mathrm{HO_2^-}] + k_7[\mathrm{HO_2}\cdot] - k_9[\mathrm{O_3}][\mathrm{O_2^-}] - k_{-7}[\mathrm{H^+}][\mathrm{O_2^-}] = 0$$

$$d[\mathrm{O_3^-}]/dt = k_9[\mathrm{O_3}][\mathrm{O_2^-}] - k_{10}[\mathrm{O_3^-}][\mathrm{H^+}] = 0$$

$$d[\mathrm{O^-}]/dt = k_8[\mathrm{OH^-}][\mathrm{HO}\cdot] - k_{11}[\mathrm{O^-}][\mathrm{H_2O}] = 0$$

Adding the five radical equations, we easily obtain:

$$2k_3[O_3][HO_2^-] - 2k_{12}[HO\cdot]^2 = 0$$

Thus, we can solve for:

$$[HO\cdot] = \sqrt{\frac{k_3[O_3][HO_2^-]}{k_{12}}}$$

CHECKPOINT 1.0 PTS

Solved: $[HO\cdot] = \sqrt{\frac{k_3[O_3][HO_2^-]}{k_{12}}}$

We know that in alkaline solutions, H_2O_2 can be considered almost completely ionized, and from the reaction:

$$H_2O_2 = HO_2^- + H^+ \quad K_2$$

we can write:

$$H_2O_2 + OH^- = HO_2^- + H_2O \quad \frac{K_2}{K_w}$$

Thus:

$$[HO_2^-] = \frac{K_2}{K_w}[H_2O_2][OH^-]$$

That is:

$$[\mathrm{HO}\cdot] = \sqrt{\frac{k_3 K_2}{k_{12} K_w}[\mathrm{O_3}][\mathrm{H_2O_2}][\mathrm{OH^-}]}$$

CHECKPOINT 1.0 PTS

Solved: $[\mathrm{HO}\cdot] = \sqrt{\frac{k_3 K_2}{k_{12} K_w}[\mathrm{O_3}][\mathrm{H_2O_2}][\mathrm{OH^-}]}$

From:

$$d[\mathrm{O_2^-}]/dt = k_3[\mathrm{O_3}][\mathrm{HO_2^-}] + k_7[\mathrm{HO_2}\cdot] - k_9[\mathrm{O_3}][\mathrm{O_2^-}] - k_{-7}[\mathrm{H^+}][\mathrm{O_2^-}] = 0$$

and

$$d[\mathrm{HO_2}\cdot]/dt = k_4[\mathrm{O_3}][\mathrm{HO}\cdot] + k_5[\mathrm{H_2O_2}][\mathrm{HO}\cdot] + k_6[\mathrm{HO_2^-}][\mathrm{HO}\cdot] - k_7[\mathrm{HO_2}\cdot] + k_{-7}[\mathrm{H^+}][\mathrm{O_2^-}] = 0$$

we can eliminate the equilibrium terms, yielding:

$$k_3[\mathrm{O_3}][\mathrm{HO_2^-}] + k_4[\mathrm{O_3}][\mathrm{HO}\cdot] + k_5[\mathrm{H_2O_2}][\mathrm{HO}\cdot] + k_6[\mathrm{HO_2^-}][\mathrm{HO}\cdot] - k_9[\mathrm{O_3}][\mathrm{O_2^-}] = 0$$

Thus, we obtain:

$$[\mathrm{O_2^-}] = \frac{k_3[\mathrm{O_3}][\mathrm{HO_2^-}] + k_4[\mathrm{O_3}][\mathrm{HO}\cdot] + k_5[\mathrm{H_2O_2}][\mathrm{HO}\cdot] + k_6[\mathrm{HO_2^-}][\mathrm{HO}\cdot]}{k_9[\mathrm{O_3}]}$$

CHECKPOINT 1.0 PTS

Eliminated terms to obtain: $[\mathrm{O_2^-}] = \frac{k_3[\mathrm{O_3}][\mathrm{HO_2^-}] + k_4[\mathrm{O_3}][\mathrm{HO}\cdot] + k_5[\mathrm{H_2O_2}][\mathrm{HO}\cdot] + k_6[\mathrm{HO_2^-}][\mathrm{HO}\cdot]}{k_9[\mathrm{O_3}]}$

Substituting our earlier results, we get:

$$[\mathrm{O_2^-}] = \frac{\frac{k_3K_2}{K_w}[\mathrm{O_3}][\mathrm{H_2O_2}][\mathrm{OH^-}] + (k_4[\mathrm{O_3}] + k_5[\mathrm{H_2O_2}] + \frac{k_6K_2}{K_w}[\mathrm{H_2O_2}][\mathrm{OH^-}])\sqrt{\frac{k_3K_2}{k_{12}K_w}[\mathrm{O_3}][\mathrm{H_2O_2}][\mathrm{OH^-}]}}{k_9[\mathrm{O_3}]}$$

CHECKPOINT 1.0 PTS

Calculated: $[\mathrm{O_2^-}] = \frac{\frac{k_3K_2}{K_w}[\mathrm{O_3}][\mathrm{H_2O_2}][\mathrm{OH^-}] + (k_4[\mathrm{O_3}] + k_5[\mathrm{H_2O_2}] + \frac{k_6K_2}{K_w}[\mathrm{H_2O_2}][\mathrm{OH^-}])\sqrt{\frac{k_3K_2}{k_{12}K_w}[\mathrm{O_3}][\mathrm{H_2O_2}][\mathrm{OH^-}]}}{k_9[\mathrm{O_3}]}$

Next, we calculate the ozone consumption rate:

$$v_{\mathrm{O_3}} = k_1[\mathrm{O_3}][\mathrm{OH^-}] + k_3[\mathrm{O_3}][\mathrm{HO_2^-}] + k_4[\mathrm{O_3}][\mathrm{HO\cdot}] + k_9[\mathrm{O_3}][\mathrm{O_2^-}]$$

Substituting the previous results, we find:

$$v_{\mathrm{O_3}} = k_1[\mathrm{O_3}][\mathrm{OH^-}] + 2\frac{K_2k_3}{K_w}[\mathrm{OH^-}][\mathrm{O_3}][\mathrm{H_2O_2}] + (2k_4[\mathrm{O_3}] + k_5[\mathrm{H_2O_2}] + \frac{K_2k_6}{K_w}[\mathrm{OH^-}][\mathrm{H_2O_2}])\sqrt{\frac{k_3K_2}{k_{12}K_w}[\mathrm{O_3}][\mathrm{H_2O_2}][\mathrm{OH^-}]}$$

CHECKPOINT 1.0 PTS

Calculated ozone consumption rate: $v_{\mathrm{O_3}} = k_1[\mathrm{O_3}][\mathrm{OH^-}] + 2\frac{K_2k_3}{K_w}[\mathrm{OH^-}][\mathrm{O_3}][\mathrm{H_2O_2}] + (2k_4[\mathrm{O_3}] + k_5[\mathrm{H_2O_2}] + \frac{K_2k_6}{K_w}[\mathrm{OH^-}][\mathrm{H_2O_2}])\sqrt{\frac{k_3K_2}{k_{12}K_w}[\mathrm{O_3}][\mathrm{H_2O_2}][\mathrm{OH^-}]}$

From the rate expression, it is evident that only option L is correct.